معهد العبور العالى
للإدارة والحاسبات ونظم المعلومات

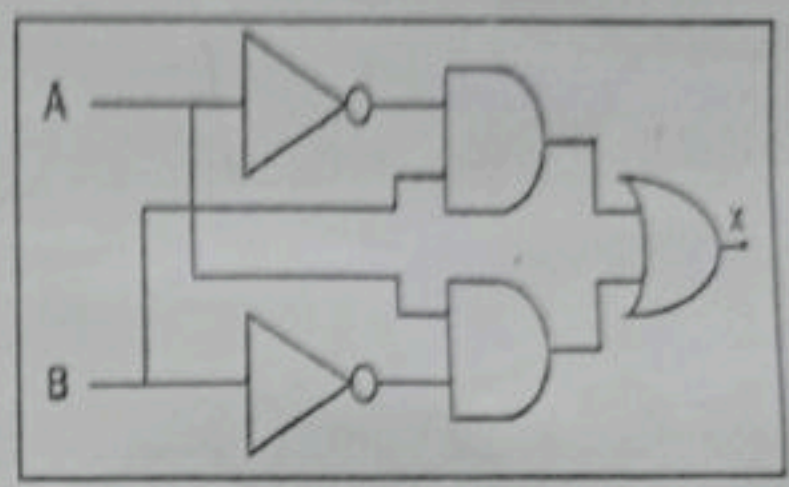
Midterm Exam of First Term (2023-2024)

Course name: Discrete Mathematical Structures	**Academic year:** 2023/2024
Course code: BAS 61101	**Date:** 19-11-2023
Year and Section: Second year-Computer Sciences	**Time:** 1 Hr. **Model(C)**
Course Instructor: Ass. Prof. Dr/ Saeed Hemeda	**No of Questions:** 3 **No of Pages:**2.

Answer the following Questions;

Question [1]: Choose the correct answer:- (8 Marks)

1) In the following Schematic diagram:



$X = \cdots$

A. $\bar{A}B$	B. $\bar{A}B + A\bar{B}$	C. $\bar{B}A + B$	D. $A\bar{B}$

2) The statement: $[\sim(p \wedge \sim q)]$ is equivalent to .....

A. $\sim p$	B. $(p \wedge \sim q)$	C. $(p \vee q)$	D. $p \rightarrow q$

3) The statement $p \rightarrow (p \vee q)$ is .....

A. $\sim p$	B. F	C. T	D. $\sim q$

4) The statement $(p \rightarrow q) \vee (p \rightarrow \sim q)$ is .....

A. T	b) F	c) q	d) $\sim p$

5) If p, q, r are three logical statements where; p is true, q is true and r is false, then the statement $[(\sim p \wedge \sim q) \vee r]$ is .....

A. $\sim r$	B. $p \wedge q$	C. 1	D. 0

6) The statement $s \rightarrow r \equiv \ldots$

A. $s \vee r$	B. $s \wedge r$	C. $\sim s \vee r$	D. $r \rightarrow s$

7) If p, q, and r is logical statement, then: $p \vee (p \wedge q)'$ is .....

A. True	B. False	C. True or false	D. False and true

8) For two sets of elements A and B ;the Boolean algebra rule: $A + \bar{A}B = \cdots$

A. $\bar{A} + B$	B. $\bar{B} + A$	C. $A + B$	D. 1

p	q	$p \wedge q$	$(p \wedge q)'$	$p \vee (p \wedge q)'$
1	1	1	0	0
1	0	0	1	1
0	1	0	1	0
0	0	0	1	0

Model (C)

معاهد العبور
Obour Institutes

وزارة التعليم العالى
معهد العبور العالى
للإدارة والحاسبات ونظم المعلومات

Question [2]: Shad the circle of T or F in the following: (8 Marks)

F 1) $p \rightarrow (q \wedge q) \equiv \sim p$.

F 2) Let's consider the interpretation v where v(p) = F, v(q) = T, v(r) = T. Then v satisfy the following propositional formula $[\sim(\sim p \rightarrow \sim q) \wedge r]$.

F 3) If $x \equiv (p \wedge q \wedge r) \vee (\acute{p} \wedge q \wedge r) \vee (p \wedge \acute{q} \wedge r) \vee (p \wedge \acute{q} \wedge \acute{r})$, then $x = \acute{q}\acute{r} + \acute{p}r$.

T 4) The family $\{\{a, b\}, \{d, c\}, \{e\}\}$ is a partitioning of the set $A = \{a, b, c, d, e\}$.

F 5) $\sim(p \wedge \sim q)$ is equivalent to $\sim p \rightarrow q$.

F 6) $q \wedge \sim(\sim p \vee q)$ is always false.

T 7) If A and B are two binary sets then: $A \triangle B = (A - B) \cup (A - B) = (A \cup B) - (A \cap B)$.

T 8) $(A - B) = A \cap B^c$.

Question [3]: Answer in details the following questions: (4 Marks)

Deduce the result of the compound statement

$$[(K \rightarrow N) \wedge (N \rightarrow S)] \rightarrow (K \rightarrow S).$$

---------- Best Wishes ..... Ass. Prof. Dr/Saeed Hemeda ----------

① P →